








Blockchain Applications

Lecture 1: Blockchains and Crypto Markets as Data



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Opening: Information in Blockchain Data for Managers

> *"The public can see that someone is sending an amount to someone else, but without information linking the transaction to anyone. This is similar to the level of information released by stock exchanges, where the time and size of individual trades, the 'tape', is made public, but without telling who the parties were."*
>
> — *Nakamoto (2008)*

- On a public blockchain every transfer records the sender address, the receiver address, the amount and the block time
  - who owns an address is not recorded; analytics firms cluster addresses into entities, with errors
- Usage: Bitcoin carried 93 112 transactions a day in January 2015 and 488 729 in September–October 2025 (30-day means)
  - Ethereum: 1.63 million a day, 3.3 times Bitcoin
- Stablecoin supply: 4.2 billion USD in January 2020, 303 billion USD on 18 September 2026
  - two issuers, Tether (USDT) and USD Coin (USDC), account for 85% of it
- Bitcoin's market value against its transactions, 132 monthly means: correlation of the logs 0.76, slope 3.6
  - in 2021–2025 the value moved without more activity

Central chart: Four Views of a Blockchain

Lecture Question and Outline

- **Question**: what can a manager learn from the data that blockchain applications produce?
  - on public blockchains (Bitcoin, Ethereum), every on-chain transfer and fee is public, carries a block time and is free to download
  - what stays hidden: owners behind addresses, trades inside exchanges, permissioned and privacy-preserving chains
- **Outline**
  - what a blockchain records, and what it does not
  - the market in numbers: coins, stablecoins, decentralised finance (DeFi), exchange-traded funds (ETFs)
  - on-chain activity: transactions, fees, security, value
  - where the data come from, and how to check them before use
  - tools for growth: log scales and growth rates
- Next lecture: the statistical behaviour of crypto prices

Learning Outcomes

- Explain what a blockchain records and why the record is public
- Name the main data sources (explorers, aggregators, dashboards) and what each measures
- Describe the size and concentration of the crypto market with market value, shares and the Herfindahl–Hirschman index (HHI)
- Read on-chain activity charts: transactions, fees, hash rate, market value
- Run a data-quality check on a crypto data set before any analysis
- Compute and interpret growth rates on a log scale

Abbreviations

AIC Akaike information criterion
AR(1) first-order autoregressive (process)
bn billion
BTC bitcoin, the coin of the Bitcoin network
CAGR compound annual growth rate
CBECI Cambridge Bitcoin Electricity Consumption Index
CC0 Creative Commons Zero (public-domain dedication)
CC BY Creative Commons Attribution licence
CRIX CRypto IndeX (Trimborn & Härdle (2018))
DeFi decentralised finance
EH/s exahashes (10^{18} hashes) per second
EODHD EOD (end-of-day) Historical Data, data vendor
ETF exchange-traded fund
ETH Ether, the coin of the Ethereum network
FABIZ Faculty of Business Administration in Foreign Languages, ASE
HHI Herfindahl–Hirschman index
IBIT iShares Bitcoin Trust, a spot Bitcoin ETF
IDA Institute Digital Assets
iid independent and identically distributed
MAD median absolute deviation
MBB Master Blockchain for Business
MFM Modelling Financial Markets (master course; Quantlet prefix)
NFT non-fungible token
S&P 500 Standard & Poor's 500 US stock index
SE standard error
SEC US Securities and Exchange Commission
SQL Structured Query Language
TVL total value locked
US United States
USD US dollar
USDC USD Coin, a stablecoin
USDT Tether, a stablecoin
UTC Coordinated Universal Time
XRP the coin of the XRP Ledger

Reading and Tools

- Core reading: Härdle, Harvey & Reule (2020), *Understanding cryptocurrencies*
- Video units on Quantinar (course Blockchain Analytics)
  - Cryptos, NFTs, Digital Assets
  - Blockchain Technology – An Introduction
  - Understanding Cryptocurrencies
- Data used today
  - daily on-chain series of Bitcoin and Ethereum, November 2014 – October 2025 (ETH Chronicles project, IDA)
  - market value, stablecoins, DeFi and ETFs up to 18 September 2026 (DefiLlama, EODHD)
- Code: the Lecture 1 notebook in the course repository reproduces every number and chart; slides built on a Modelling Financial Markets (MFM) Quantlet link it at the bottom right

Recap: Information Recorded on the Ledger

- Hashing and consensus were covered in earlier courses; here: what each piece produces as data
- **Ledger**: a public blockchain keeps one public list of transactions, copied on thousands of computers (**nodes**)
  - permissioned chains, privacy mechanisms and off-chain trades are not in this public record
- **Block**: a batch of transactions with a timestamp; it carries the **hash** (fingerprint) of the previous block, so old records are tamper-evident
  - the block time is when the transfer was recorded, not the exact time of the economic exchange
- **Consensus**: the rule that decides who adds the next block
  - proof of work (Bitcoin): difficulty and hash rate are in the data (next slide)
  - proof of stake (Ethereum since 15 September 2022): the stake is in the data; energy use fell by more than 99.9% ethereum.org: The Merge
- **Smart contract**: a program stored on the chain; its calls (deposits, loans, swaps) are on-chain data
- **Layer 2**: a network that bundles many transactions and settles them on Ethereum; only the settlement reaches the main chain

Proof of Work in Numbers

- A block is valid if the hash of its header with a number (**nonce**) is below a target T:

$$H(\text{header} \,\|\, \text{nonce}) < T, \qquad H \sim \text{Uniform}\{0, \ldots, 2^{256} - 1\}$$

- One try succeeds with probability $p = T/2^{256}$; tries until success are geometric:

$$\mathrm{E}[\text{tries}] = \frac{1}{p} = \frac{2^{256}}{T} = D \cdot 2^{32}, \qquad D = \frac{T_{\max}}{T}, \; T_{\max} \approx 2^{224}$$

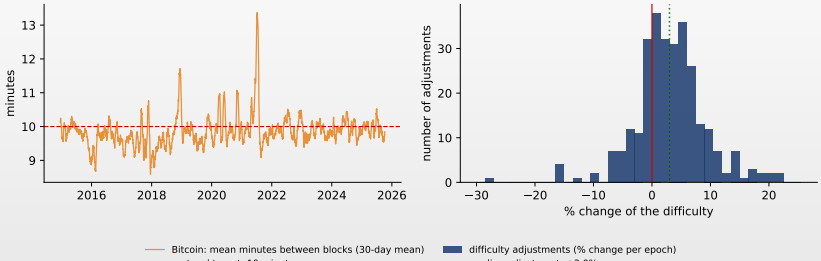
  - D: the **difficulty**, published in every block

- One block every 600 seconds on average, hence the network **hash rate**

$$\hat{H} = \frac{D \cdot 2^{32}}{600} \text{ hashes per second}$$

  - 8 October 2025: $D = 1.508 \times 10^{14}$, so $\hat{H} = 1.508 \times 10^{14} \times 4.295 \times 10^{9}/600 = 1.08 \times 10^{21} = 1\,080$ EH/s (exahashes per second)
  - every 2 016 blocks: $D_{\text{new}} = D_{\text{old}} \times 20\,160$ min/actual minutes

Block Intervals and the 10-Minute Target



- Left: daily mean time between Bitcoin blocks, 30-day mean, November 2014 – 8 October 2025; right: the 291 difficulty adjustments in the data (% change at each new epoch of 2 016 blocks)
- Mean time between blocks 9.86 minutes: 146.1 blocks a day against the target of 144; below 10 minutes on 59% of the days
- Median adjustment +3.0%; 74% of the adjustments go up: the hash rate keeps growing, blocks come early, and the difficulty catches up
- Largest fall −27.9% (epoch starting 4 July 2021, after China's mining ban; China's share of the hash rate fell to zero in July 2021, CBECI mining map); the 30-day mean block time reached 13.4 minutes that month

BA_L1_blocktime

Worked Example: One Difficulty Adjustment

- Rule: every 2 016 blocks, $D_{\text{new}} = D_{\text{old}} \times 20\,160/m$, with m the minutes the last 2 016 blocks took
- Data: $D_{\text{old}} = 1.4234 \times 10^{14}$ (epoch from 19 September 2025), $D_{\text{new}} = 1.5084 \times 10^{14}$ (from 2 October 2025)
- Step 1: factor $D_{\text{new}}/D_{\text{old}} = 1.5084/1.4234 = 1.0597$; the difficulty rises by 5.97%
- Step 2: solve for the minutes, $m = 20\,160/1.0597 = 19\,024$ minutes $= 13.21$ days instead of 14
- Step 3: mean time between blocks $= 19\,024/2\,016 = 9.44$ minutes
  - daily mean block times give 9.52 minutes for 19 September – 1 October 2025; days do not start and end with the epoch
- Reading: the new difficulty brings the same hash rate back to 10 minutes per block; if the hash rate keeps growing, the next epoch is early again

The Supply Rule

- Block subsidy at block height h: $s(h) = 50 \cdot 2^{-\lfloor h/210\,000 \rfloor}$ BTC
  - halved every 210 000 blocks, about four years: 50, 25, 12.5, 6.25, 3.125 (April 2024)
- Total supply, a geometric series:

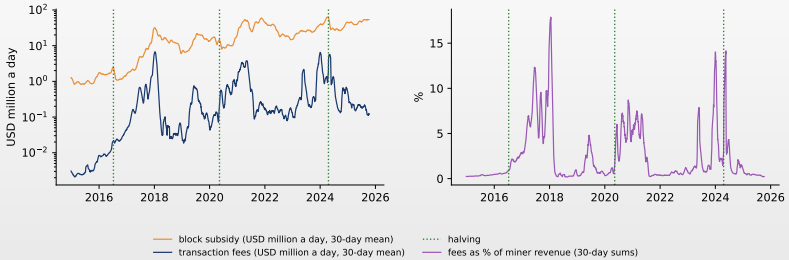
$$S_\infty = 210\,000 \sum_{k=0}^{\infty} 50 \cdot 2^{-k} = 210\,000 \times 50 \times 2 = 21 \text{ million BTC}$$

- Annual **supply growth rate** after the 2024 halving:

$$g_S = \frac{3.125 \times 144 \times 365}{S_t} = \frac{164\,250}{20.09 \times 10^6} = 0.82\%$$

  - 144 blocks a day; $S_t = 20.09$ million BTC (coins in circulation) on 18 September 2026
  - growth of the coin stock, not consumer-price inflation: not comparable with a central bank's 2% inflation target
- The supply path is known in advance: it cannot explain price *surprises*

Miner Revenue: Block Subsidy and Fees



- **Miner revenue** per day = subsidy (blocks × subsidy × price) + fees (transactions × average fee); on-chain series to 8 October 2025, EODHD closes
- Fees as a share of miner revenue: 10.4% in 2017, 4.0% in 2023, 0.4% in 2025; highest month December 2017 (15.7%)
- 30 days to 8 October 2025: fees are 0.23% of revenue; miners earn 53.7 million USD a day, almost all of it from the subsidy
- Each halving cuts the subsidy in BTC by half: the security budget depends on the price, and on fees once the subsidy is small

BA_L1_fee_share ▶ Interactive

Worked Example: One Day of Miner Revenue

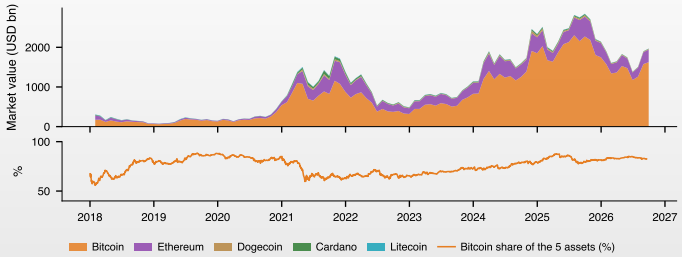
- Bitcoin, 8 October 2025
- Step 1, blocks: mean block time 9.73 minutes, so $1\,440/9.73 = 148.0$ blocks
- Step 2, subsidy: $148.0 \times 3.125 = 462.5$ BTC; at the close of 123 355 USD, 57.1 million USD
- Step 3, fees: 427 326 transactions $\times$ 0.350 USD $=$ 0.150 million USD
- Step 4, fee share: $0.150/(57.1 + 0.150) = 0.26\%$
- After the next halving (subsidy 1.5625 BTC), with the same price and fees
  - revenue falls from 57.2 to 28.7 million USD a day
  - the fee share doubles to 0.52%: still far from replacing the subsidy

On-Chain and Off-Chain Information

On-chain (public blockchains)	Off-chain (private or partial)
every transfer: sender address, receiver address, amount, time	who owns an address: names are not recorded
fees paid, block size, block time	trades inside an exchange (one internal ledger)
supply of each coin and token	reserves backing a stablecoin
smart-contract calls: deposits, loans, swaps	prices on exchanges (each venue has its own)
mining or staking power (hash rate, stake)	the business purpose of a transaction

- Addresses are **pseudonymous**: address $\neq$ entity (an exchange, a fund) $\neq$ economic owner (the client); analytics firms cluster addresses into entities, with errors
- Most trading happens off-chain, on centralised exchanges: on-chain data show settlement, not all trading

Market Value of Five Selected Coins and Bitcoin's Share



- **Market value** of coin i: $M_{i,t} = P_{i,t} \times Q_{i,t}$, the price in USD times the coins in circulation
  - five selected coins with a public circulating supply (BTC, ETH, DOGE, ADA, LTC), not the five largest
  - together: 1 970 billion USD on 18 September 2026; peak 3 135 billion on 6 October 2025
  - Bitcoin: 1.63 trillion USD, 82.5% of this five-coin basket (55.8% to 88.5% since 2018); not its share of the whole market
- The value moves with prices; supply changes slowly

MFM_ch16_market_value ▶ Interactive

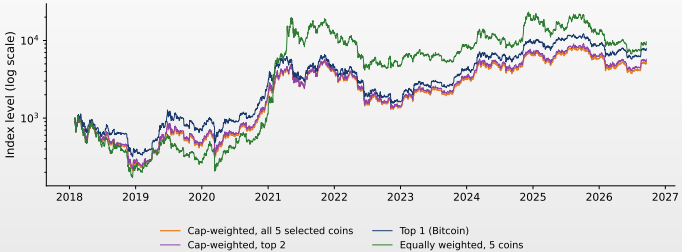
Concentration of the Five-Coin Basket

- **Herfindahl–Hirschman index** (HHI), the measure competition authorities use for market concentration

$$\mathrm{HHI} = \sum_{i=1}^{N} w_i^2, \qquad w_i = \frac{M_i}{\sum_j M_j}$$

  - w_i: the value share of coin i; N: the number of coins
  - from $1/N$ (equal shares) to 1 (one coin holds everything)
  - $1/\mathrm{HHI}$: the **effective number** of coins
- **Worked example**: shares 82.5%, 15% and 2.5%
  - $\mathrm{HHI} = 0.825^2 + 0.15^2 + 0.025^2 = 0.681 + 0.023 + 0.001 = 0.70$
  - effective number $1/0.70 = 1.4$: these shares behave like 1.4 equal coins
- The five selected coins (BTC, ETH, DOGE, ADA, LTC) on 18 September 2026: $\mathrm{HHI} = 0.71$, effective number 1.42
  - a measure for this basket, not for the whole market; adding coins lowers it (eleven coins, September 2025: 0.58, Appendix)

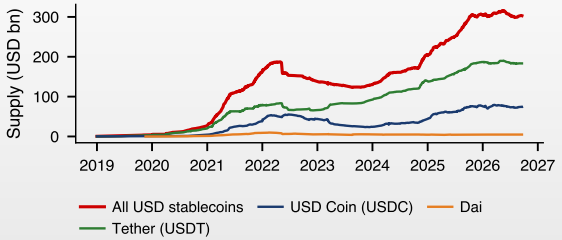
A Market Index: CRIX



- **CRIX** (CRypto IndeX) Trimborn & Härdle (2018): a value-weighted index of the largest coins; the number of coins is chosen by a statistical rule
- Chart: the same construction on the five selected coins, monthly rebalancing, 1 000 on 31 January 2018
- Five-coin index over $T = 8.6$ years: compound annual growth rate (CAGR) $(I_T/I_0)^{1/T} - 1 = 21.4\%$; annualised volatility of daily returns 65.7%
- **Survivorship bias**: an index of today's coins ignores the coins that died; past returns look better than an investor could have earned
- Lecture 3: the full construction, the real CRIX data, its volatility index VCRIX, and nine other types of crypto indices

MFM_ch16_crix ▶ Interactive

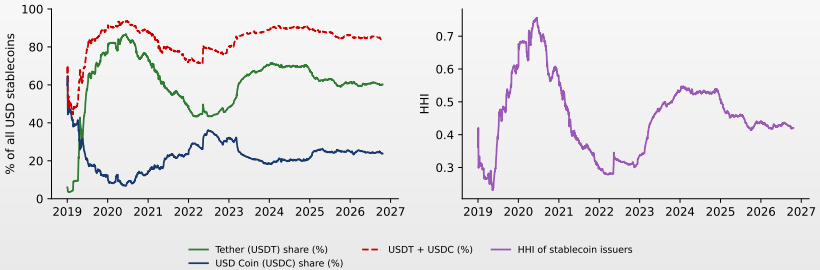
Growth of Stablecoin Supply



- **Stablecoin**: a token designed to keep a fixed price, usually 1 USD (Lectures 5 and 6)
- Supply (DefiLlama): 4.2 billion USD in January 2020, 303 billion on 18 September 2026, about 73 times more
  - Tether (USDT) 183 billion (61%), USD Coin (USDC) 74 billion (24%): two issuers hold 85%
  - 2022–2023 dip: from 187 to 123 billion, after TerraUSD collapsed and USDC lost its peg

MFM_ch16_stablecoins ▶ Interactive

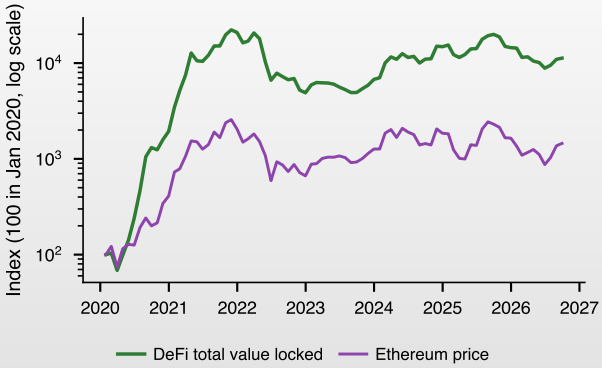
Stablecoin Issuers



- Shares of all USD stablecoins (DefiLlama, daily, 2019–2026) and the HHI of issuers
- Tether: 76.8% (January 2020), 44.0% (April 2022), 67.5% (August 2023), 60.5% (September 2026); USD Coin: 12.4%, 27.9%, 20.7%, 24.4%
- HHI 0.607, 0.282, 0.500, 0.427: concentration fell during the 2021–2022 boom and rose again after the USDC depeg of March 2023 (Lecture 5)

BA_L1_stablecoin_conc ▶ Interactive

DeFi: Total Value Locked



- **Total value locked** (TVL): the value of tokens deposited in DeFi protocols (DefiLlama)
- 0.6 billion USD in January 2020, 177 billion at the peak (9 November 2021), 88 billion today
- **Caution**: TVL mixes deposits with token prices (Appendix)

MFM_ch16_defi ▶ Interactive

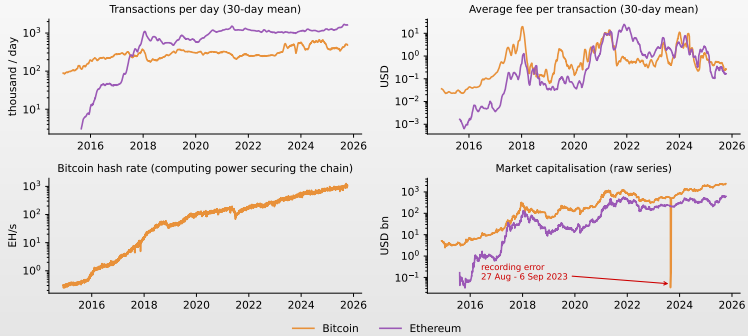
January 2024: Spot Bitcoin Exchange-Traded Funds



- **Spot ETF**: an exchange-traded fund that holds the coins themselves SEC (2024)
  - 10 January 2024: 11 spot Bitcoin ETFs approved in the US
- Chart: Bitcoin and the ETF IBIT, both 100 on 11 January 2024; the two lines overlap
- An off-chain application of an on-chain asset: pension funds can now hold Bitcoin through a regulated security

MFM_ch16_etf ▶ Interactive

Four Views of a Blockchain



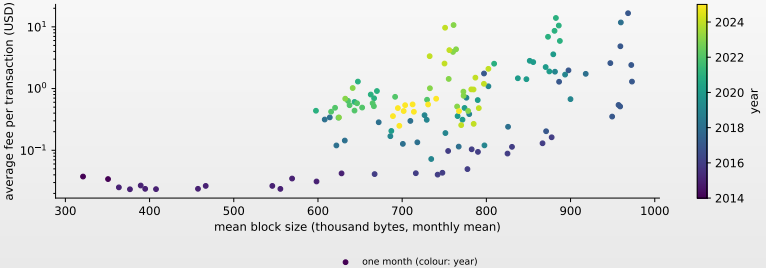
- Daily series, November 2014 – 8 October 2025, log scales; data: ETH Chronicles project (IDA)

BA_L1_onchain ▶ Interactive

Reading the Four Panels

- **Usage**: Bitcoin 93 112 transactions a day in January 2015, 488 729 in September–October 2025 (30-day mean)
  - Ethereum: 1.63 million a day, 3.3 times Bitcoin; one possible reason: smart-contract use generates many small transactions
- **Fees**: average fee per transaction in USD, 30-day mean
  - peaks in congestion: 19.81 USD (Bitcoin, January 2018), 24.79 USD (Ethereum, November 2021)
  - October 2025: 0.27 USD (Bitcoin) and 0.17 USD (Ethereum)
  - a possible explanation for Ethereum: activity moved to layer-2 networks; the chart alone does not identify the cause
- **Security**: the Bitcoin hash rate rose from 0.30 to 1 079 EH/s (exahashes per second), about 3 600 times
  - the cost of an attack rose too, but not in proportion: hardware became more efficient and energy prices changed
- **Value**: Bitcoin 2 442 billion USD, Ethereum 620 billion on 8 October 2025

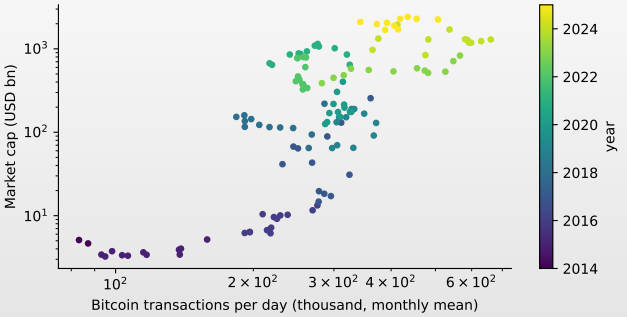
Block Utilisation and Transaction Fees



- One point per month, 132 months: Bitcoin mean block size (thousand bytes) and average fee per transaction (USD, log scale)
- Correlation of log fee with log block size 0.65; with log transactions per day 0.63
- Quarter of months with the fullest blocks (at least 813 thousand bytes): median fee 1.88 USD; emptiest quarter (at most 664): 0.32 USD
- Block space is limited, so users bid for it when demand is high (December 2017: 968 thousand bytes, 16.63 USD); an association of monthly means, not a demand curve

BA_L1_congestion

Market Value and Network Activity



- Monthly means, 132 months; both axes on log scales
- Correlation of the logs: 0.76
- Slope of the log-log line: 3.6
  - 1% more transactions goes with about 3.6% more market value
- Network-value theories (Metcalfe's law) predict a slope of 2
- But the cloud is wide: in 2021–2025 value moved without more activity
- Two trending series can look related: Lecture 2 examines this

BA_L1_value_activity ▶ Interactive

Where the Data Come From

Source	What it measures	Use in this course
Etherscan, mempool.space	block explorers: single transactions, addresses, contracts	check one transaction or one wallet
CoinGecko, EODHD, Yahoo Finance	prices, volumes and market value aggregated over exchanges	returns, volatility, correlations (L2, L4)
DefiLlama	stablecoin supply, DeFi TVL, fees by protocol	stablecoins (L5, L6), DeFi
Dune	Structured Query Language (SQL) queries on decoded blockchain tables	custom on-chain metrics
Cambridge Bitcoin Electricity Consumption Index (CBECI)	electricity use of Bitcoin mining	background: the energy cost of proof of work

- All five are free for daily series
- Sources differ in exchanges covered, close time and definitions: the same metric can differ across providers
- Providers revise and relabel series: keep a dated copy of every series analysed, so that results can be reproduced

Four Typical Errors in On-Chain Data

- On-chain series are scraped, merged and relabelled by data providers; four kinds of error are common

Error	Example in the lecture data	How to detect it	What to do
Recording error	Bitcoin market value 1 000 times too small on 11 days, 27 August – 6 September 2023	plot on a log scale	set the days to missing
Mislabelled series	the "transaction value" file equals the "transaction fee" file	compare series pairwise	do not use the series
Series that ends early	social-media counts stop on 14 March 2023	last non-missing date	end the sample there
Missing values	XRP transactions: 32% missing since 2020, none after November 2023	share missing per year	drop the coin from that series

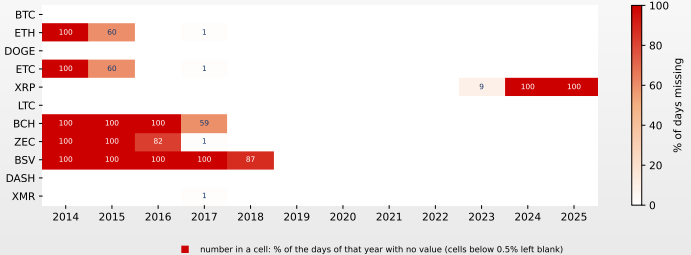
- None of these errors stops a program from running; each biases a mean, a correlation or a growth rate without any warning

A Checklist Before Any Analysis

1. **Calendar**: one row per day? duplicate dates? gaps? (Bitcoin, 28 November 2014 – 8 October 2025: 3 968 days, no gaps)
2. **Coverage**: first and last valid date of every series
3. **Missing values**: the share per series and per year
4. **Units**: USD or coins? billions or units? per day or cumulative?
5. **Outliers**: plot every series on a log scale; jumps of 100 times are errors, not news
6. **Duplicates**: two series that are identical, or a series copied into the wrong column
7. **Source and definition**: which exchanges, which close time, which address set

- Steps 2, 3, 5 and 6 detect the four typical errors of the previous slide

Coverage and Missing Values, Coin by Coin



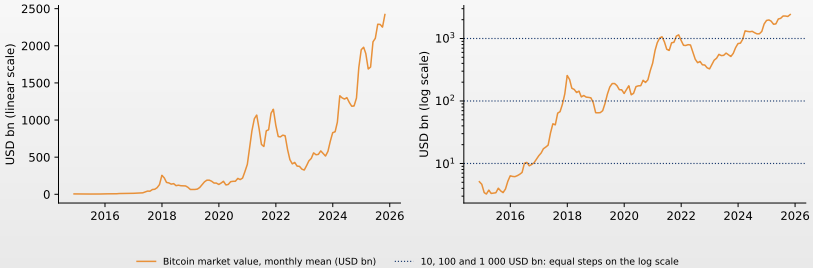
- Share of days with no value in the daily transaction counts, 11 coins by year (2014 from 28 November, 2025 to 8 October)
  - ETC Ethereum Classic, BCH Bitcoin Cash, ZEC Zcash, BSV Bitcoin SV, XMR Monero
- Most gaps come before a coin's first value: BCH starts on 1 August 2017, BSV on 15 November 2018
- After its first value a coin misses 0–0.2% of the days, except XRP: 17.2%, with no value at all in 2024–2025
- All 11 coins are observed only from 15 November 2018 to 27 November 2023: a balanced panel is five years, not eleven

BA_L1_missing

Log Scales for Exponential Growth

- Crypto series grow by factors of 100 or 1 000: on a linear axis the early years are a flat line
- On a **log scale**, equal distances mean equal **ratios**
  - 1 to 10 and 100 to 1 000 have the same height: both are a 10-fold rise
  - a straight line means a constant growth rate
- **Log return** between two dates: $r = \ln(X_{\text{end}}/X_{\text{start}})$
  - log returns add up over time: the return of a year is the sum of its daily log returns
  - for small changes, $r \approx$ the percentage change
- Rule: plot levels on a log scale, analyse log returns (Lecture 2)

One Series on a Linear and a Log Scale



- Bitcoin market value, monthly means, November 2014 – October 2025 (October: 1–8 October), the 11 erroneous days of 2023 excluded
- Linear scale (left): January 2015, 3.4 billion USD, is 0.14% of the October 2025 value of 2 421 billion; 54% of the months sit below a tenth of that value and look flat
- Log scale (right): 10, 100 and 1 000 billion USD are equal steps; the fall from 256 (December 2017) to 65 billion (December 2018), −75%, is as visible as the later booms

BA_L1_logscale

Worked Example: Log Returns Add Up

- Bitcoin market value, monthly means: 256.1 (December 2017), 64.8 (December 2018), 403.1 billion USD (December 2020)
- Simple returns: $64.8/256.1 - 1 = -74.7\%$, then $403.1/64.8 - 1 = +522.2\%$
  - their sum, $+447.5\%$, is wrong: the two-year change is $403.1/256.1 - 1 = +57.4\%$
- Log returns: $\ln(64.8/256.1) = -1.375$, then $\ln(403.1/64.8) = +1.828$
  - their sum, $+0.454$, equals $\ln(403.1/256.1)$
  - back to a percentage change: $e^{0.454} - 1 = +57.4\%$
- Rule: add log returns over time; convert to a percentage change only at the end

Worked Example: the Growth Rate of Bitcoin Usage

- **Compound annual growth rate** (CAGR) between two dates T years apart

$$g = \left(\frac{X_{\text{end}}}{X_{\text{start}}} \right)^{1/T} - 1$$

- Bitcoin transactions, monthly means: $X_{\text{start}} = 93\,112$ (January 2015), $X_{\text{end}} = 506\,379$ (September 2025), $T = 10.67$ years
  - ratio $= 5.44$; $\ln 5.44 = 1.694$; $1.694/10.67 = 0.159$; $g = e^{0.159} - 1 = 17.2\%$ a year
- Bitcoin market value, same months: 3.4 and 2 251 billion USD
  - ratio $= 659$; $g = 659^{1/10.67} - 1 = 83.8\%$ a year: value grew much faster than usage
- Each CAGR depends on the two dates chosen; the next slide uses every month

The Log-Linear Trend Regression

- A CAGR uses only two dates; a trend regression uses all of them:

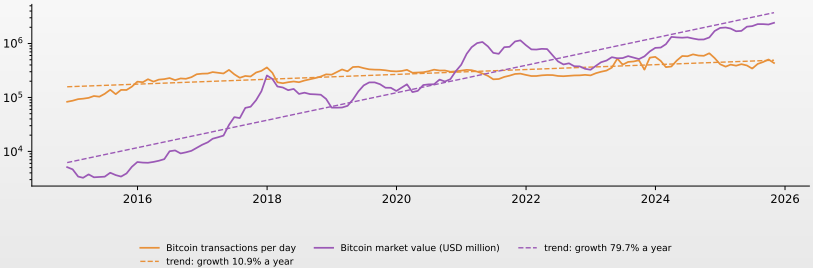
$$\ln X_t = a + g\,t + u_t, \qquad t \text{ in years}$$

  - g: the continuous growth rate; annual growth $e^{g} - 1$
  - SE: standard error of $\hat{g}$, Newey–West (robust to autocorrelation, 12 lags); R^2: share of the variance of $\ln X_t$ explained by the trend

Bitcoin, monthly, 2014–2025	$\hat{g}$	SE	R^2	$e^{\hat{g}} - 1$
transactions per day	0.104	0.021	0.62	10.9%
market value	0.586	0.055	0.88	79.7%

- Difference, estimated directly from $\ln(M_t/n_t) = a + (g_M - g_n)\,t + e_t$ (M_t: market value; n_t: transactions per day): 0.482 (SE 0.052), $z = 9.3$
  - but the residual AR(1) coefficient is 0.96: deviations from the trend persist for years; the test is only as good as the trend model
- The trend uses every month, unlike the CAGRs (17.2% and 83.8%), but it still depends on the sample period

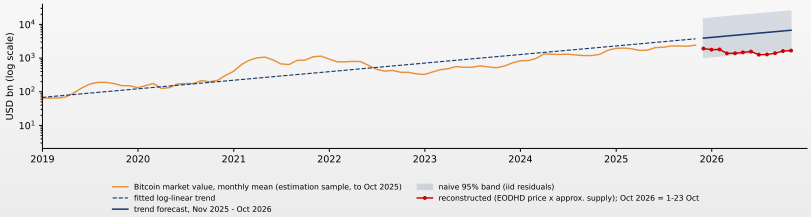
Trend Lines on a Log Scale



- Monthly means and fitted log-linear trends: transactions grow 10.9% a year, market value 79.7% a year
- On a log scale a constant growth rate is a straight line; the market value moves around its trend in long waves (2017–2018, 2021–2022)

BA_L1_trend ▶ Interactive

Out-of-Sample Check of the Trend after One Year



Bitcoin market value, USD bn	Nov 2025	Feb 2026	Jun 2026	Sep 2026	Oct 2026*
trend forecast (79.7% a year)	3 918	4 537	5 516	6 387	6 706
reconstructed (EODHD price × approx. supply)	1 920	1 376	1 263	1 617	1 670
reconstructed / forecast	0.49	0.30	0.23	0.25	0.25

* **1–23 October 2026. Supply from the protocol rule at the block rate observed since 2009 (Appendix), not an observed monthly series.**

- One year on, the reconstructed market value is a quarter of the trend; it ends just below the naive 95% band
- Residuals are almost a random walk (first-order autoregressive (AR(1)) coefficient 0.97): the independent and identically distributed (iid) band is far too narrow beyond a few months; a trend fits the past, it does not forecast a price

BA_L1_oos_trend ▶ Interactive

Summary

- A blockchain is a public, tamper-evident database; its security can be measured: $\hat{H} = D \cdot 2^{32}/600 \approx 1\,080$ EH/s for Bitcoin in October 2025
- Supply follows a known geometric rule: 21 million BTC, supply growth 0.82% a year after 2024
- Concentration: $\mathrm{HHI} = 0.71$ for five selected coins (BTC, ETH, DOGE, ADA, LTC); two issuers hold 85% of stablecoins
- Log-linear trends, 2014–2025: Bitcoin usage grows 10.9% a year, market value 79.7% a year; in this sample the gap is large ($z = 9.3$), but residuals are very persistent
- Data must be checked before use: several prices for one asset, gaps, and jumps of 100 times that are errors, not news
- Levels and trends describe the past; inference and forecasts need returns: Lecture 2 turns the same prices into returns

Interpretation Exercises

1. Five selected coins (BTC, ETH, DOGE, ADA, LTC) have $\mathrm{HHI} = 0.71$. A colleague says the crypto market is well diversified because thousands of coins exist. Comment, using the effective number of coins. What would you need to measure the concentration of the whole market? (Lecture 3 repeats the measure on ten coins.)
2. Trend regressions: $\hat{g} = 0.104$ for transactions, 0.586 for market value; the difference is 0.482 (SE 0.052, $z = 9.3$), with residual AR(1) coefficient 0.96. What does the test tell you? Does it prove a bubble?
3. The Bitcoin hash rate grew about 3 600 times since 2014. What does this say about the cost of an attack, and what does it not say about the value of the network?
4. The 79.7% trend predicted 6 706 bn USD for October 2026; the reconstructed value (1–23 October) is 1 670 bn. The residual AR(1) coefficient is 0.97. Why was the forecast so far off, and why is the naive band misleading?

⊡ Watch Blockchain Mechanism and Distributional Characteristics of Cryptos

Appendix: Further Material

Optional material for self-study.

- Worked Example: a Miner with 1% of the Hash Rate
- Checking the Hash-Rate Formula on the Data
- The Supply Rule in Calendar Time
- Building a Market-Value Index
- Worked Example: the Divisor at a Rebalancing
- Share of TVL Explained by the Ether Price
- Worked Example: TVL Without New Deposits
- A Robust Outlier Rule
- The Outlier Rule on the Data
- Bitcoin and Ethereum: Milestones, 2008–2026
- The Supply Schedule
- Coins, Tokens and Smart Contracts
- Concentration Through Time
- Where the Value Is Locked
- Price Differences across Exchanges and the Daily Close

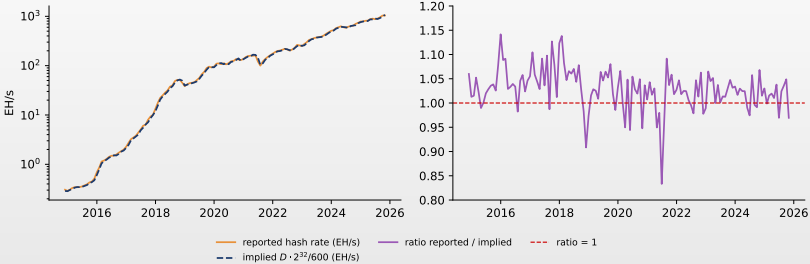
Worked Example: a Miner with 1% of the Hash Rate

- Each block is won by the miner with probability equal to its share of the hash rate, independently of the past
- Blocks won in a day by a miner with share $s = 1\%$: approximately Poisson with mean $\lambda = 144\, s = 1.44$

$$P(N = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \qquad P(N = 0) = e^{-1.44} = 0.237, \qquad P(N \geq 3) = 0.176$$

- Business reading
  - on almost one day in four, a miner with 1% of the network earns nothing, although its costs (electricity) are fixed
  - revenue variance is the reason for **mining pools**: miners share hash rate and receive a smoothed payout
- The same model explains why exchanges wait for several **confirmations**: the chance that an attacker with share $q < 0.5$ catches up falls geometrically with the number of blocks

Checking the Hash-Rate Formula on the Data



- Monthly means, November 2014 – October 2025: reported hash rate against $D \cdot 2^{32}/600$ from the published difficulty
- Ratio reported/implied: mean 1.029, standard deviation 0.042, range 0.83–1.14
- The reported series is itself an estimate from block times: the 3% gap reflects blocks arriving slightly faster than every 600 seconds

BA_L1_hashrate_check

The Supply Rule in Calendar Time

- At exactly 144 blocks a day, the supply rule gives 19 992 600 BTC on 24 October 2026
- Actual supply (EODHD, 24 October 2026): 20 102 728 BTC, i.e. 110 128 BTC (0.55%) more
- Block height implied by the actual supply, from the fourth epoch (subsidy 3.125 BTC since height 840 000):

$$h = 840\,000 + \frac{20\,102\,728 - 19\,687\,500}{3.125} = 972\,873$$

- Actual rate since the first block: 972 873/6 503 days = 149.6 blocks a day, 3.9% faster than the target
  - The difficulty adjusts only every 2 016 blocks; while the hash rate grows, blocks come early
- Next halving at height 1 050 000: at 144 blocks a day about 11 April 2028
- Lesson: the *rule* is exact in blocks, not in calendar time; the time path is a forecast with a small, one-sided error

Building a Market-Value Index

- Value-weighted index of k coins with a **divisor** D_t:

$$I_t = \frac{\sum_{i=1}^{k} P_{i,t}\, Q_{i,t}}{D_t}$$

- At a rebalancing date t^* the constituents change; the divisor keeps the index continuous:

$$D_{t^*}^{\text{new}} = D_{t^*}^{\text{old}} \cdot \frac{\sum_{i \in \text{new}} P_{i,t^*}\, Q_{i,t^*}}{\sum_{i \in \text{old}} P_{i,t^*}\, Q_{i,t^*}}$$

- How many coins? Trimborn & Härdle (2018) choose k by an Akaike information criterion (AIC)-type rule (stylised form)

$$\hat{k} = \arg\min_{k} \ \mathrm{AIC}(k) = n \ln \hat{\sigma}_k^2 + 2k$$

  - $\hat{\sigma}_k^2$: the variance of the difference between the k-coin index return and the total-market return
  - more coins track better but cost more to replicate; AIC balances the two

Worked Example: the Divisor at a Rebalancing

- Day t^*, before rebalancing: coins A and B, market values 800 and 200 (USD bn); divisor $D = 1$, index $I = 1\,000$
- Coin C (market value 100) enters the index at t^*

$$D^{\text{new}} = 1 \times \frac{800 + 200 + 100}{800 + 200} = 1.1, \qquad I_{t^*} = \frac{1\,100}{1.1} = 1\,000$$

  - the index does not jump: a change of constituents is not a return
- Next day: A 820, B 190, C 110, total 1 120

$$I_{t^*+1} = \frac{1\,120}{1.1} = 1\,018.2, \qquad \text{return } +1.82\%$$

- Without the divisor update the index would have shown a false $+10\%$ on the rebalancing day

Share of TVL Explained by the Ether Price

- Monthly regression of log changes, Newey–West standard errors (SE):

$$\Delta \ln \mathrm{TVL}_t = a + b\,\Delta \ln P_t^{\mathrm{ETH}} + u_t$$

  - b: the elasticity of TVL to the Ether price
- Estimates, 80 months: $\hat{b} = 0.79$ (SE 0.066), $R^2 = 0.62$
  - $t = 12.0$: the association is precisely estimated
  - $R^2 = 0.62$: Ether price changes account statistically for 62% of the variance of monthly TVL changes
- Decomposition: $\mathrm{TVL}_t = \sum_j P_{j,t}\, N_{j,t}$, prices times token quantities deposited
  - activity is better measured with fixed-price quantities $\sum_j \bar{P}_j N_{j,t}$, net flows or fees

MFM_ch16_defi

Worked Example: TVL Without New Deposits

- A lending protocol holds 100 million USD: 60% in Ether, 40% in stablecoins
- Ether rises by 50%; nobody deposits or withdraws

$$\mathrm{TVL}_{t+1} = 60 \times 1.5 + 40 = 130 \text{ million USD}, \qquad +30\%$$

- In this accounting example, the elasticity of TVL to the Ether price is 0.6: the Ether share of the deposits
- Fixed-price TVL $\sum_j \bar{P}_j N_{j,t}$ stays at 100: activity did not change
- The market-wide $\hat{b} = 0.79$ is a different object: an association between monthly percentage changes of TVL and of the Ether price, 2020–2026
  - it mixes the asset composition of deposits, flows that move with Ether and other token prices; it is not the Ether share of deposits

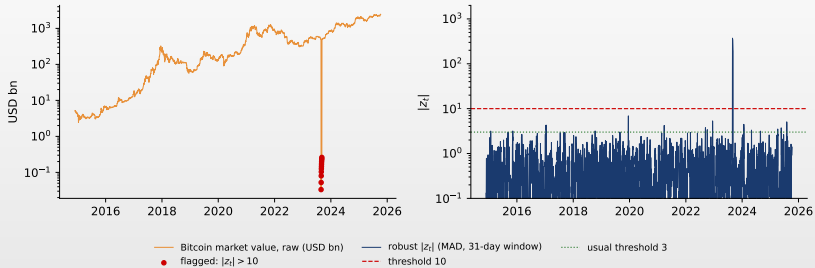
A Robust Outlier Rule

- Mean and standard deviation are themselves distorted by outliers; use the median and the **median absolute deviation** (MAD)

$$z_t = \frac{x_t - \mathrm{med}_t}{1.4826\,\mathrm{MAD}_t}, \qquad \mathrm{MAD}_t = \mathrm{med}_{s \in W_t} \left| x_s - \mathrm{med}_t \right|$$

  - x_t: the log market value; W_t: the centred 31-day window; med_t: the median of x_s on W_t
  - MAD_t is computed in each window, around that window's own median
  - 1.4826: makes $1.4826\,\mathrm{MAD}$ equal to σ for Normal data
- Bitcoin market value, 2014–2025
  - the 11 days of 27 August – 6 September 2023 have $|z_t|$ between 57 and 373
  - the largest genuine move has $|z_t| = 6.9$ (17 December 2019): a threshold of 10 flags exactly the 11 bad days
  - a threshold of 100 would miss 2 of them (the first two days of the error)
- A rule written down before looking at new data is better than deleting points by eye

The Outlier Rule on the Data

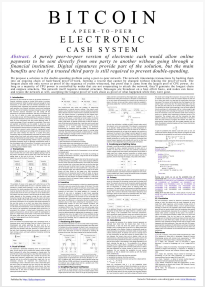


- Left: raw Bitcoin market value with the flagged days; right: robust $|z_t|$ on a log scale
- The usual threshold $|z_t| > 3$ flags 46 of 3 968 days, 35 of them genuine moves: heavy tails make genuine moves look extreme
- Threshold 10 flags exactly the 11 bad days; any threshold between 6.9 and 57 separates the two groups in this sample

BA_L1_mad

Bitcoin and Ethereum: Milestones, 2008–2026

- 31 October 2008: Nakamoto (2008) proposes "a peer-to-peer electronic cash system"
  - **peer-to-peer**: users pay each other directly, without a bank in the middle
  - the problem solved: **double spending**, i.e. paying twice with the same digital coin
- 3 January 2009: the first block; 2010: the first trades on exchanges
- 2015: Ethereum adds **smart contracts**, programs that run on the blockchain Buterin (2014)
- 2026: stablecoins, tokenised funds, DeFi and exchange-traded funds are all built on these two ideas



BITCOIN

A PEER-TO-PEER

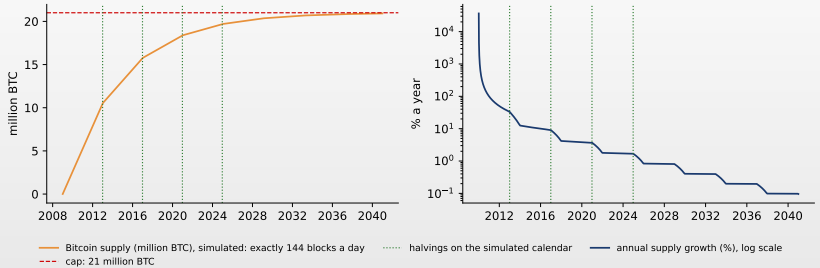
ELECTRONIC

CASH SYSTEM

The Bitcoin white paper, 2008

Text: Satoshi Nakamoto (2008); poster: DailyCoinPost (2018); CC0; Wikimedia Commons

The Supply Schedule

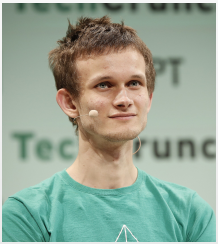


- Simulated calendar: the protocol rule with exactly 144 blocks a day; dotted green: the four halvings on this calendar (end of 2012, 2016, 2020, 2024)
- The actual halvings came earlier (November 2012, July 2016, May 2020, April 2024): blocks arrived faster than every 10 minutes
- Annual supply growth: 33.3% (2013), 9.0% (2017), 3.7% (2021), 1.7% (2025), 0.83% (2026), 0.4% (2032); 99% of the cap is reached in 2035
- The simulation gives 19.98 million BTC on 18 September 2026; the actual supply is 20.09 million

BA_L1_supply_schedule

Coins, Tokens and Smart Contracts

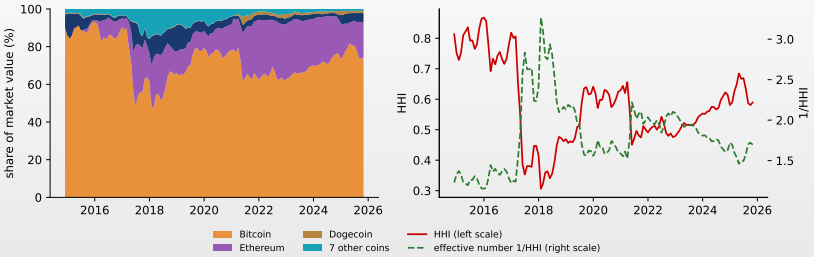
- **Coin**: the native asset of a blockchain (BTC on Bitcoin, ETH on Ethereum)
  - pays the fees; Bitcoin supply is capped at 21 million by the protocol
- **Token**: an asset issued by a smart contract on an existing chain
  - stablecoins (USDT, USDC), governance tokens, tokenised funds and bonds
- **DeFi** (decentralised finance): lending and trading run by smart contracts, without a bank or broker
- Each of these is an **application**; each leaves a data trail on the chain



Vitalik Buterin, co-founder of Ethereum, 2015

Photo: John Phillips (2015); CC BY 2.0; Wikimedia Commons

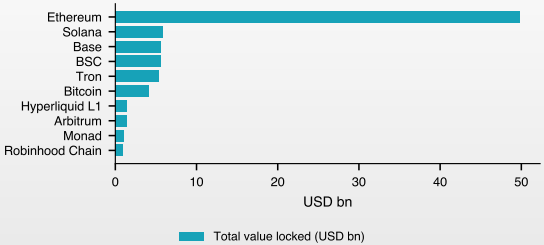
Concentration Through Time



- Monthly market-value shares of 11 coins (ETH Chronicles data, 2014–2025; the 11 erroneous days of 2023 excluded)
- HHI: 0.868 at its maximum (December 2015); 0.306 at its minimum (January 2018, Bitcoin share 47.1%, effective number 3.27)
- September 2025: HHI 0.581, Bitcoin share 73.5%, effective number 1.72: the market re-concentrated after each altcoin boom

BA_L1_hhi_coins

Where the Value Is Locked



- TVL by blockchain, ten largest, 18 September 2026: Ethereum 49.8 billion USD, 57% of 88 billion
  - Base and Arbitrum are **layer-2** networks (Recap slide): cheaper transactions that settle on Ethereum
- Business reading: one platform dominates, but applications spread to cheaper chains

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Price Differences across Exchanges and the Daily Close

- Hundreds of exchanges, each with its own order book and price
  - a data provider averages them into one series; the choice of venues changes the series
- Makarov & Schoar (2020): 34 exchanges in 19 countries, 2017–2018
  - large and persistent price gaps between countries (Korea, Japan, US) when capital controls block arbitrage
  - much smaller gaps within a country
- The market never closes: 365 trading days a year
  - a "daily close" is a convention, usually midnight UTC (Coordinated Universal Time)
  - the S&P 500 closes at 16:00 New York time: daily returns of the two are not simultaneous